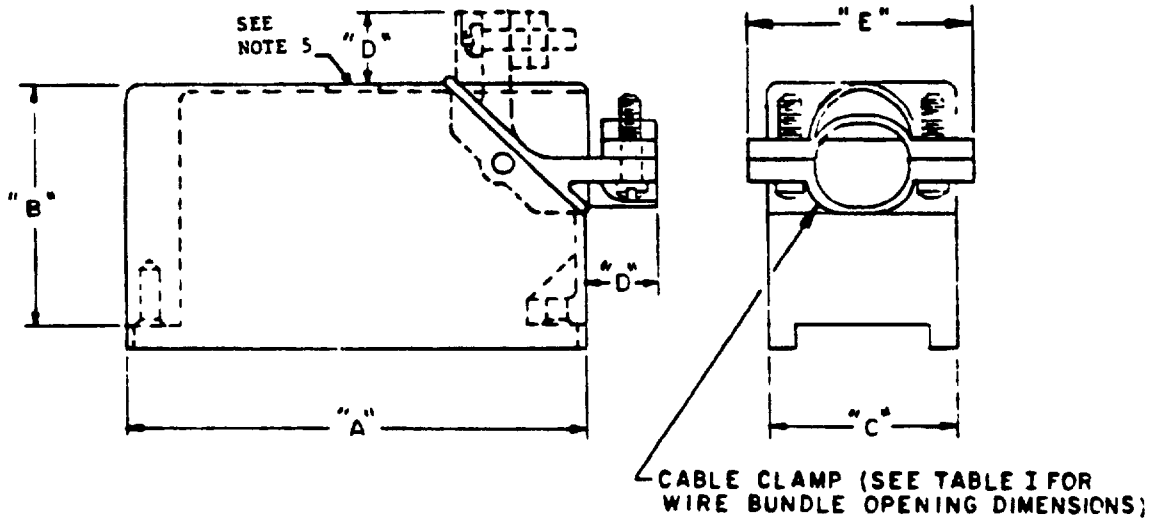
MILITARY SPECIFICATION SHEET

# CONNECTOR, ELECTRICAL, RECTANGULAR SHIELD, FOR 20, 38, 56, 90 OR 120 REMOVABLE CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-28731.



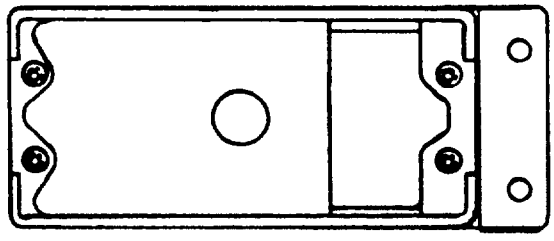



FIGURE 1. Shield, rectangular.

Ⓓ denotes changes



FSC 5935

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Shield assemblies can be used with the following connector assemblies, M28731/9-0109 and -0012; M28731/10-0109, -0111, -0112 and -0114; M28731/11-0073 and -0075; M28731/12-0073 and -0075; M28731/13-0073 and -0075, M28731/14-0073 and -0075; M28731/15-0073 and -0075; M28731/16-0073 and -0075, M28731/17-0109 and -0112; M28731/18-0109, -0111, -0112 and -0114, M28731/29-0001 and -0002; M28731/30-0001 and -0002, M28731/31-0001 and -0002, M28731/32-0001 and -0002, M28731/33-0001 and -0002, M28731/34-0001 and -0002.
4. Shield assemblies are supplied with (4) .086-56, .122-40, .138-32 screws for mounting shield to insulator. Complete unit to be packaged in accordance with MIL-P-116, method 111.
5. Shield assemblies M28731/19-0011 through -0014 and -0019 through -0022 are supplied with a rubber grommet No. MS-35489-4 per MIL-G-3036 for placement in clearance hole of acturating hardware shaft.
6. Patent Notice. Patent No. 3,248,686 owned by Elco corporation. The government has a royalty-free license under the above patent for the benefit of manufacturers of the items called for in this specification, either for the government or for use in equipment to be delivered to the government.

FIGURE 1. Shield, rectangular - Continued.



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(D) TABLE I. Part number information.

| Military part number shield | For use with connector specification sheet | Dim 'A' Max | Dim 'B' ± 020 | Dim 'C' Max | Dim 'D' Max | Dim 'E' Max | Clamp location | Wire bundle opening Ref. |
|---|---|---|---|---|---|---|---|---|
| M28731/19-0001<br>M28731/19-0002 | MIL-C-28731/9 and /10 | 1.340 | .756 | .720 | .340 | .720 | Side<br>Top | .454 Dia. |
| M28731/19-0003<br>M28731/19-0004 | MIL-C-28731/11, /12, /29 and /30 | 2.025 | 1.381 | .920 | .500 | .920 | Side<br>Top | .650 x .500 |
| M28731/19-0005<br>M28731/19-0006 | MIL-C-28731/11, /12, /29 and /30 | 2.025 | 1.381 | .920 | .500 | 1.110 | Side<br>Top | .650 x .608 |
| M28731/19-0007 1/<br>M28731/19-0008 1/ | MIL-C-28731/13, 14, /31 and /32 | 2.618 | 1.381 | .920 | .500 | .920 | Side<br>Top | .650 x .500 |
| M28731/19-0009<br>M28731/19-0010 | MIL-C-28731/13, /14, /31 and /32 | 2.618 | 1.381 | .920 | .500 | 1.110 | Side<br>Top | .650 x .608 |
| M28731/19-0011<br>M28731/19-0012 | MIL-C-28731/15, /16, /33 and /34 | 2.745 | 1.625 | 1.260 | .500 | 1.260 | Side<br>Top | .800 Dia. |
| M28731/19-0013<br>M28731/19-0014 | MIL-C-28731/17 and /18 | 2.778 | 1.875 | 1.560 | .500 | 1.560 | Side<br>Top | 1.080 x .800 |
| M28731/19-0015<br>M28731/19-0016 | MIL-C-28731/11, /12, /29 and /30 | 2.025 | 1.381 | .920 | .500 | 1.110 | Side<br>Top | .820 x .614 |
| M28731/19-0017<br>M28731/19-0018 | MIL-C-28731/13, /14, /31 and /32 | 2.618 | 1.381 | .920 | .500 | 1.110 | Side<br>Top | .820 x .614 |
| M28731/19-0019<br>M28731/19-0020 | MIL-C-28731/15, /16, /33 and /34 | 2.745 | 1.625 | 1.260 | .500 | 1.260 | Side<br>Top | .800 x 1.000 |
| M28731/19-0021<br>M28731/19-0022 | MIL-C-28731/17 and /18 | 2.778 | 1.875 | 1.560 | .500 | 1.560 | Side<br>Top | 1.100x1.085 |

1/ Shield will only accommodate wire sizes 22, 24, and 26 AWG. Use shield M28731/19-0009 and M28731/19-0010 for wire sizes 18 and 20 AWG.



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REQUIREMENTS

Dimensions and configuration  See figure 1 and table I.

Part number  See table I.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, MI
Navy - AS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AL, AV, ME
Navy - MC, OS
Air Force - 19

Agent
DLA - ES

Preparing activity
Navy - EC

(Project 5935-3345-17)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1 DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-C-28731/19D | Connector, Electrical, Rectangular Shield, For 20, 38, 56, 90 or 120 Removable Contacts |

| 3a NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
| b ADDRESS *(Street, City, State, ZIP Code)* | |

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a NAME OF SUBMITTER *(Last, First, MI)* – Optional | b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8 DATE OF SUBMISSION *(YYMMDD)* |

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